

Lecture 06: Review for Midterm 1

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Learning Objectives

Electron as a particle

- ▶ Drude model and Hall effect
- ▶ Maxwell's equations
- ▶ Calculate ω_p and δ

Wave-Particle duality

- ▶ de Broglie wavelength
- ▶ Heisenberg uncertainty
- ▶ Group vs phase velocity

Quantum physics

- ▶ Solve Schrödinger equation
- ▶ ψ , k , E , ω , and boundary conditions
- ▶ Example potentials (infinite and finite wells, barrier)

Hydrogen atom

- ▶ Coulomb potential and spherical symmetry
- ▶ Radial probability distribution
- ▶ Energy levels

Drude model: from drift velocity to conductivity

Key assumptions

Electrons undergo random thermal motion with average velocity reset after each collision with the crystal (time τ)

Derivation:

- ▶ Drift velocity acquired between collisions:

$$v_D = -\frac{eE\tau}{m} \quad (1)$$

- ▶ Current density from carrier motion:

$$J = -N_e e \cdot v_D = \frac{N_e e^2 \tau}{m} E \quad (2)$$

- ▶ Conductivity is:

$$\sigma = \frac{N_e e^2 \tau}{m} = N_e e \mu \quad (3)$$

Insight: Validates Ohm's law $J = \sigma E$ at microscopic level

Drude model: typical values and connections

Microscopic quantities:

- ▶ Collision time: $\tau \sim 10^{-14}$ s
- ▶ Drift velocity: $v_D \sim 10^{-3}$ m/s
- ▶ Thermal velocity: $v_{th} \sim 10^5$ m/s
- ▶ Mean free path: $\ell = v_{th}\tau$

Mobility:

$$\mu = \frac{e\tau}{m} = \frac{v_D}{E} \quad (4)$$

In experiments

Macroscopic σ measurement gives microscopic τ :

$$\tau = \frac{m\sigma}{N_e e^2} \quad (5)$$

But what about N_e ?

Note: $v_D \ll v_{th}$ means drift is a small perturbation on random thermal motion

Hall effect: carrier characterization

Force balance in the steady-state:

- ▶ Lorentz force: $F_B = ev_D B$
- ▶ Hall electric force: $F_E = eE_H$
- ▶ Equilibrium: $eE_H = ev_D B$

Hall coefficient:

$$R_H = \frac{E_H}{J_x B} = \frac{1}{N_e e} \quad (6)$$

Key properties:

- ▶ Depends ONLY on carrier density
- ▶ Independent of mobility and mass
- ▶ Sign reveals carrier type

From experiments

From measured Hall voltage (V_H), current (I), magnetic field (B), and thickness (t):

$$R_H = \frac{V_H \cdot t}{I \cdot B} \quad (7)$$

$$N_e = \frac{1}{R_H \cdot e} \quad (8)$$

$$\mu = \sigma R_H \quad (9)$$

Plasma frequency and skin depth in metals

Plasma frequency from the Maxwell's equations

$$\omega_p = \sqrt{\frac{N_e e^2}{m \epsilon_0}} \quad (10)$$

Frequency response:

- ▶ $\omega < \omega_p$: metal opaque ($\epsilon_{eff} < 0$)
- ▶ $\omega > \omega_p$: metal transparent ($\epsilon_{eff} > 0$)
- ▶ typical metals: $\omega_p \sim 2 \times 10^{15}$ Hz (UV)

Skin depth:

$$\delta = \sqrt{\frac{2}{\omega \mu \sigma}} \quad (11)$$

- ▶ Field decay: $E(z) = E_0 e^{-z/\delta}$
- ▶ Cu at 2.4 GHz: $\delta \approx 1.33 \mu\text{m}$

Physical meaning: ω_p is the natural oscillation frequency of the electron gas

Worked example: Drude conductivity calculation

Problem

Copper has $N_e = 8.5 \times 10^{28} \text{ m}^{-3}$ and $\sigma = 5.9 \times 10^7 \text{ S/m}$. Calculate the collision time τ and drift velocity v_D in a field $E = 1 \text{ V/m}$.

Step 1: Find collision time from conductivity

$$\tau = \frac{m\sigma}{N_e e^2} = \frac{(9.11 \times 10^{-31})(5.9 \times 10^7)}{(8.5 \times 10^{28})(1.6 \times 10^{-19})^2} \quad (12)$$

$$\tau = 2.5 \times 10^{-14} \text{ s} \quad (13)$$

Step 2: Calculate drift velocity

$$v_D = \frac{eE\tau}{m} = \frac{(1.6 \times 10^{-19})(1)(2.5 \times 10^{-14})}{9.11 \times 10^{-31}} \quad (14)$$

$$v_D = 4.3 \times 10^{-3} \text{ m/s} \approx 4.3 \text{ mm/s} \quad (15)$$

Check: $v_D \ll v_{th} \sim 10^6 \text{ m/s}$ ✓

Checkpoint: classical electron models

Conceptual questions

1. Write the Drude conductivity formula. What does each symbol represent?
2. Why does the Hall coefficient depend only on carrier density?
3. How to obtain the wave equation from the Maxwell's equations?
4. What happens to EM waves in a metal below the plasma frequency?
5. Can you write down the assumptions of the Drude model? What about the Hall effect? What about the plasma frequency?

Wave-particle duality: de Broglie relation

de Broglie relation

$$\lambda = \frac{h}{p} = \frac{h}{mv} \quad (16)$$

Bridges particle momentum to wave wavelength

Experimental validation: Davisson-Germer

- ▶ Electrons accelerated through voltage V gain kinetic energy eV
- ▶ Momentum: $p = \sqrt{2m \cdot eV}$
- ▶ Bragg diffraction from crystal: $n\lambda = d \sin \theta$
- ▶ Two independent measurements of λ agree!

Key insight

Wave nature observable when $\lambda \sim$ system size (atomic spacing ~ 0.1 nm)

Higher momentum $\rightarrow$ shorter wavelength $\rightarrow$ more particle-like

Heisenberg uncertainty principle: fundamental limits

Position-Momentum:

$$\Delta p \cdot \Delta z \geq \hbar/2 \quad (17)$$

Atomic stability

If electron collapses to nucleus:

- ▶ $\Delta z \rightarrow 0$
- ▶ $\Delta p \rightarrow \infty$
- ▶ Kinetic energy $\rightarrow \infty$

Uncertainty prevents collapse!

Energy-Time:

$$\Delta E \cdot \Delta t \geq \hbar/2 \quad (18)$$

NOT a measurement limitation!

Intrinsic property of wave nature

Clarification

Heisenberg originally proposed $\Delta p \cdot \Delta z \sim \hbar$, just an order of magnitude estimate. Sometimes, you will see discrepancies on the uncertainty principle, though $\hbar/2$ is the lower bound for the uncertainty.

Group vs phase velocity: energy transport

Phase velocity:

$$v_p = \frac{\omega}{k} \quad (19)$$

Motion of constant-phase surfaces (wave crests)

Group velocity:

$$v_g = \frac{\partial \omega}{\partial k} \quad (20)$$

Motion of wave packet envelope (“particle-like” entity)

Remember!

- ▶ Energy travels at v_g
- ▶ Information travels at v_g
- ▶ v_p can exceed c !
- ▶ For electrons: $v_g = v_{classical}$

Dispersion relation for a free electron:

$$\omega = \frac{E}{\hbar} = \frac{\hbar k^2}{2m} \Rightarrow v_g = \frac{\hbar k}{m} = \frac{p}{m} = v_{particle} \quad (21)$$

Checkpoint: wave-particle duality

Quick Check Questions

1. Why is the uncertainty principle fundamental, not a measurement issue?
2. Which velocity (group or phase) carries energy?
3. How does confinement affect minimum kinetic energy?
4. How would you demonstrate the uncertainty principle without having to resort to quantum mechanics?
5. How do you define a “particle”, in the end?

Schrödinger equation

Time-dependent Schrödinger equation (postulate)

$$-\frac{\hbar^2}{2m}\nabla^2\Psi + V\Psi = i\hbar\frac{\partial\Psi}{\partial t} \quad (22)$$

Separation of variables: $\Psi(\mathbf{r}, t) = \psi(\mathbf{r}) \cdot T(t)$ when V is time-independent

Time-independent:

$$\boxed{-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi} \quad (23)$$

Time evolution:

$$T(t) = e^{-iEt/\hbar} \quad (24)$$

Just a phase change

Eigenvalue equation for energy E

Stationary states: $|\Psi(\mathbf{r}, t)|^2 = |\psi(\mathbf{r})|^2$ is time-independent

Boundary conditions and step potential

At finite potential steps: ψ and $\frac{d\psi}{dz}$ must be **continuous**

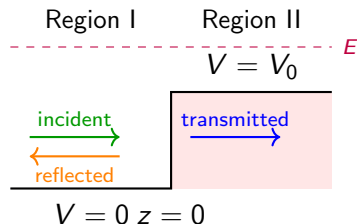
Potential Step ($E > V_0$):

- ▶ Region I: $k_1 = \sqrt{2mE/\hbar^2}$
- ▶ Region II: $k_2 = \sqrt{2m(E - V_0)/\hbar^2}$
- ▶ Note: $k_2 < k_1$

Boundary Conditions at $z = 0$:

$$\psi_1(0) = \psi_2(0) \quad (25)$$

$$\left. \frac{d\psi_1}{dz} \right|_0 = \left. \frac{d\psi_2}{dz} \right|_0 \quad (26)$$



Quantum tunneling

Classically forbidden region ($E < V$)

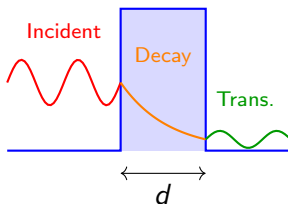
Wavefunction decays exponentially, but is NOT zero!

Decay Constant:

$$\gamma = \sqrt{\frac{2m(V - E)}{\hbar^2}} \quad (27)$$

Wavefunction Inside Barrier:

$$\psi = Ae^{-\gamma z} + Be^{+\gamma z} \quad (\text{real exponentials, not oscillatory}) \quad (28)$$



Infinite potential well: energy quantization

Boundary Conditions: $\psi(0) = \psi(L) = 0$ at rigid walls

Solution inside well ($V = 0$):

$$\psi_n(z) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi z}{L}\right) \quad (29)$$

Quantized wave vectors:

$$k_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots \quad (30)$$

Quantized energies:

$$E_n = \frac{h^2 n^2}{8mL^2} = \frac{\hbar^2 \pi^2 n^2}{2mL^2} \quad (31)$$

Key Features

- ▶ $E_1 \neq 0$
- ▶ $E_n \propto n^2$ (not equally spaced)
- ▶ $E \propto 1/L^2$ (confinement)
- ▶ Infinite bound states

Finite potential well

Inside well: Oscillatory

$$\psi \propto \sin(kz), \quad k = \frac{\sqrt{2mE}}{\hbar} \quad (32)$$

Outside well: Exponential decay

$$\psi \propto e^{-\gamma z}, \quad \gamma = \frac{\sqrt{2m(V_1 - E)}}{\hbar} \quad (33)$$

From boundary condition:

$$\boxed{k \tan\left(\frac{kL}{2}\right) = \gamma} \quad (34)$$

Compared to infinite well

- ▶ No analytical solution
- ▶ Solve graphically/numerically
- ▶ **Finite** number of bound states
- ▶ Wavefunction penetrates barriers
- ▶ No need to solve the transcendental equation in exams

Insight on energy: Deeper/wider wells support more bound states; very shallow wells may have only one bound state

Worked example: infinite well energy levels

Problem

An electron is confined in a quantum well of width $L = 1$ nm. Calculate: (a) Ground state energy E_1 , (b) First excited state E_2 , (c) Photon wavelength for $2 \rightarrow 1$ transition.

Step 1: Ground state energy ($n = 1$)

$$E_1 = \frac{h^2}{8mL^2} = \frac{(6.63 \times 10^{-34})^2}{8(9.11 \times 10^{-31})(10^{-9})^2} = 6.03 \times 10^{-20} \text{ J} \quad (35)$$

$$E_1 = 0.377 \text{ eV} \quad (36)$$

Step 2: First excited state ($n = 2$)

$$E_2 = 4E_1 = 1.51 \text{ eV} \quad (37)$$

Step 3: Transition wavelength

$$\Delta E = E_2 - E_1 = 3E_1 = 1.13 \text{ eV} = 1.81 \times 10^{-19} \text{ J} \quad (38)$$

$$\lambda = \frac{hc}{\Delta E} = \frac{(6.63 \times 10^{-34})(3 \times 10^8)}{1.81 \times 10^{-19}} = 1.10 \text{ } \mu\text{m (infrared)} \quad (39)$$

Checkpoint: postulates of quantum mechanics

Conceptual questions

1. What boundary conditions apply at a finite potential step? Why?
2. Why is there partial reflection even when $E > V_0$?
3. How does tunneling probability depend on barrier width?
4. Why is it useful to know that energies are quantized upon confinement?

Hydrogen atom: ground state and Bohr radius

Ground state wavefunction

$$\psi_{1s} \propto e^{-r/a_0} \quad (40)$$

Bohr radius:

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2} = 52.8 \text{ pm} \quad (41)$$

Ground State Energy:

$$E_1 = -13.6 \text{ eV} \quad (42)$$

Ionization energy of hydrogen

Key Insight

a_0 is the **most probable radius**, NOT an orbit!

- ▶ $|\psi|^2$ maximum at $r = 0$
- ▶ $r^2|\psi|^2$ maximum at $r = a_0$
- ▶ $\langle r \rangle = 1.5a_0$

Worked example: hydrogen atom transitions

Problem

Calculate the wavelength of light emitted when a hydrogen atom transitions from $n = 3$ to $n = 2$ (H_α line of Balmer series).

Step 1: Calculate energy levels

$$E_3 = -\frac{13.6}{3^2} = -\frac{13.6}{9} = -1.51 \text{ eV} \quad (43)$$

$$E_2 = -\frac{13.6}{2^2} = -\frac{13.6}{4} = -3.40 \text{ eV} \quad (44)$$

Step 2: Energy of emitted photon

$$\Delta E = E_3 - E_2 = -1.51 - (-3.40) = \boxed{1.89 \text{ eV}} \quad (45)$$

Step 3: Convert to wavelength

$$\lambda = \frac{hc}{\Delta E} = \frac{1240 \text{ eV} \cdot \text{nm}}{1.89 \text{ eV}} = \boxed{656 \text{ nm (red light)}} \quad (46)$$

Note: Used $hc = 1240 \text{ eV} \cdot \text{nm}$ for quick calculation

Checkpoint: hydrogen atom

Conceptual questions

1. What is the physical meaning of the Bohr radius?
2. Why does hydrogen energy depend only on n ?
3. How does one compute probability values in different systems of coordinates?
4. How to derive the solution to the hydrogen atom from the Schrödinger equation?
5. How is this different from the Bohr's model?